

Euclid's Elements — Book II

Proposition 1

Formal Reconstruction — Technical Documentation

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Given two straight lines, if one is divided into several segments, then the rectangle contained by the two is equal to the sum of the rectangles contained by the undivided line and the several parts.

1 Introduction

Proposition II.1 opens Book II with a change of mathematical language. Book I has developed a synthetic theory of congruence, parallels, parallelograms, squares, and equal content. Book II begins to combine these ingredients in a systematic calculus of rectangles. A modern reader immediately recognizes in II.1 the pattern of a distributive law. That recognition is useful, but it must be stated carefully: Euclid does not introduce variables for numerical lengths, does not define multiplication of lengths, and does not prove an identity in a number field.

The primitive object in the proposition is a geometric figure: a rectangle “contained by” two straight lines. When one of the two lines is divided into parts, the large rectangle is divided into smaller rectangles having the same other side. The equality asserted by the proposition is therefore first of all a statement about decomposition of figures.

This is a natural point at which to speak of the beginning of the *geometrical algebra* characteristic of Book II. The phrase is an interpretive description of a stable pattern of geometric operations, not a claim that Euclid was secretly manipulating modern symbolic formulas. In the present formal reconstruction this distinction is especially sharp: the code contains rectangles, segment congruence, betweenness, parallelograms, right angles, and scissors relations, but no multiplication operation on segment lengths and no numerical area function.

The formal development separates three levels:

1. a concrete rectangle and its geometric cuts;
2. the Euclidean phrase “rectangle contained by” two given segments;
3. independence of the particular rectangle chosen to represent that phrase.

The final theorem is stronger than a bare equal-content statement. Its conclusion is the exact relation

HilbertScissorsEq

for the whole rectangle and the formal sum of its parts.

2 Euclid’s “Rectangle Contained By”

2.1 The original geometric meaning

Book II begins with the terminology that a rectangular parallelogram is “contained by” the two straight lines which contain one of its right angles. Thus the expression does not initially denote a product of two numbers. It names a rectangle by two adjacent sides.

Suppose a rectangle has adjacent sides congruent to two given segments PQ and RS . Euclid’s phrase

the rectangle contained by PQ, RS

should be read geometrically: choose a right-angled parallelogram whose two adjacent sides have those magnitudes. The rectangle itself remains a plane figure.

For exposition only, we will occasionally write

$\text{Rect}(PQ, RS)$.

This is documentary shorthand. It is not a Lean definition, it is not a number-valued function, and the comma is not multiplication.

2.2 Why this already looks algebraic

If BC is divided into consecutive parts BM , MN , and NC , II.1 says in this shorthand

$$\text{Rect}(BC, PQ) = \text{Rect}(BM, PQ) + \text{Rect}(MN, PQ) + \text{Rect}(NC, PQ).$$

A modern algebraist naturally sees the shape

$$(b_1 + b_2 + b_3)a = b_1a + b_2a + b_3a.$$

The formal proof, however, establishes the geometric statement on the first line, not the numerical identity on the second. The second formula is only a modern mnemonic for the combinatorial pattern of the dissection.

This is consistent with the modern synthetic reading of Euclid’s area theory. Hartshorne treats Book II as a collection of equal-content statements about figures and describes the resulting manipulation as “geometrical algebra.” The point is precisely that the manipulation precedes any assignment of numerical areas to the figures.

3 The Synthetic Rectangle Layer

3.1 Rectangles

The reconstruction introduces a small reusable rectangle API in `HilbertRectangle.lean`. A rectangle is a parallelogram with one specified right angle:

```
def IsRectangle
  (A B C D : Geo.Point) : Prop :=
  IsParallelogram Geo A B C D /\
  HilbertRightAngle Geo A B C
```

The parallelogram structure supplies the opposite-side relations, while the right-angle component distinguishes rectangles from arbitrary parallelograms. The remaining right angles can then be propagated synthetically.

3.2 The formal meaning of “contained by”

The Euclidean phrase is represented by the predicate

```
def IsRectangleContainedBy
  (A B C D P Q R S : Geo.Point) : Prop :=
  IsRectangle Geo A B C D /\
  Geo.Congruent A B P Q /\
  Geo.Congruent B C R S
```

Thus

```
IsRectangleContainedBy Geo A B C D P Q R S
```

means exactly that $ABCD$ is a concrete rectangle, $AB \cong PQ$, and $BC \cong RS$. Nothing in this definition assigns a number to PQ or RS . The two given segments specify the two adjacent geometric magnitudes of a representative rectangle.

The layer also proves existence of such a representative and, more importantly for the semantics of Book II, uniqueness of its scissors content. The theorem

```
rectangle_contained_by_unique
```

says that any two rectangles contained by the same ordered pair of segments satisfy the exact scissors relation. Consequently Euclid’s definite description “the rectangle contained by PQ, RS ” becomes well-defined at the scissors level even though many different concrete quadrilaterals can represent it.

3.3 No segment multiplication and no numerical area

There are two tempting modernizations that the present reconstruction explicitly avoids.

First, we do not define a product of segment lengths $|PQ| |RS|$. The pair (PQ, RS) remains a pair of geometric segments, and the rectangle is related to that pair by congruence of adjacent sides.

Second, we do not map a rectangle to a real number called its area. The content of the proposition is expressed through relations between finite formal sums of triangles. This permits addition, dissection, transport, and comparison of figures at the synthetic level.

This is also conceptually distinct from Hilbert's later arithmetic of segments, where arithmetic operations are introduced as an additional construction. No such field or segment-arithmetic layer is imported into II.1.

4 Exact Scissors Equality and Equal Content

The project uses two related but different content relations. They should not be conflated.

4.1 Exact scissors equality

`HilbertScissorsEq` records an exact finite cut-and-paste relation between formal scissors terms. In II.1 it is produced by explicit splits of the rectangle into smaller rectangles and by normalization of the triangulated parallelogram terms. Schematically,

$$P \simeq_{\text{sc}} Q + R$$

means that the two sides can be matched by the exact scissors machinery already developed in the project.

The rectangle transport theorem is already available in this stronger form:

```
rectangle_transport_scissorsEq
```

for rectangles with congruent corresponding adjacent sides.

4.2 The more general equal-content relation

The broader project notion corresponding to equal content is

```
HilbertScissorsEquicomplementable
```

It allows suitable complementary terms to be adjoined before exact scissors equality is established. Thus it is designed to capture a more general synthetic equal-content relation.

The theorem

```
rectangle_transport
```

is obtained by weakening the exact transport theorem from `HilbertScissorsEq` to equicomplementability.

For II.1 no such weakening is necessary. The final theorem proves `HilbertScissorsEq` itself. Therefore II.1 certainly yields an equal-content consequence, but its formal statement retains the stronger fact that the displayed equality comes from an exact dissection.

5 The Classical and Formal Configurations

5.1 The binary split

The reusable core is a one-cut theorem. Let $B - M - C$ and let the corresponding point L divide the opposite side DE . The rectangle $BCED$ is cut into the rectangles $BMLD$ and $MCEL$:

$$BCED \simeq_{\text{sc}} BMLD + MCEL.$$

In Lean the theorem is

```
rectangle_split
```

and it returns both the rectangle structure of the two pieces and the exact scissors equality.

The formal configuration contains one additional point X . It satisfies $D - X - C$ and $M - X - L$. This is not decorative data: the inherited exact split lemma from the I.47 scissors infrastructure uses the intersection of the cut line with a diagonal. The classical rectangle picture suppresses this witness; the formal theorem makes it explicit.

5.2 The three-part split

The three-part form applies the binary split twice. First cut at $M - L$; then cut the remaining rectangle at $N - K$. The result is

$$\begin{aligned} BCED \simeq_{\text{sc}} & BMLD \\ & + MNKL \\ & + NCEK. \end{aligned}$$

The second split has its own diagonal-intersection witness Y , with $L - Y - C$ and $N - Y - K$.

Figure 1 shows both levels and includes the hidden witnesses X and Y required by the current Lean interface.

The actual three-part theorem is

```
rectangle_split_three
```

Its proof is structurally transparent: obtain the first binary equality, split the remaining right-hand rectangle, add the unchanged left piece to that second equality, and compose by transitivity of `HilbertScissorsEq`.

5.3 Scope of the present formal statement

Euclid states II.1 for one line cut into an arbitrary finite number of parts. The current Lean file does not yet package an n -ary list-valued theorem. It proves the binary split and the explicit

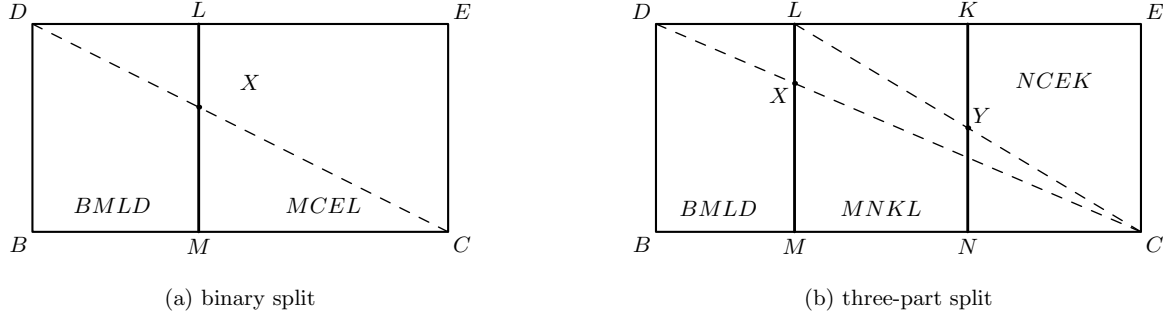


Figure 1: The exact rectangle splits underlying II.1. Panel (a) is the binary kernel: $B - M - C$ and the cut ML divide $BCED$ into $BMLD$ and $MCEL$; the dashed diagonal DC meets the cut at the formal witness X . Panel (b) applies the same construction twice: BC is divided into BM , MN , and NC , producing the three rectangles $BMLD$, $MNKL$, and $NCEK$. The dashed diagonals display the witnesses X and Y used by the two exact split calls.

three-part form used in the standard II.1 configuration. The binary theorem is the reusable kernel from which a general finite version could later be obtained by iteration.

This is a deliberate statement-fidelity point. The documentation records Euclid’s general formulation, but the final formal statement displayed below is the representative-independent three-part theorem actually present in `Proposition2_1.lean`.

6 From a Concrete Diagram to Euclid’s Language

The geometric split alone produces concrete rectangles. Euclid II.1 is more abstract: it speaks of the rectangles *contained by* the whole segment and the several parts together with one fixed segment.

The theorem

```
rectangle_split_three_contained_by
```

bridges these levels. If $CE \cong PQ$, it packages the four concrete rectangles as

$$\begin{aligned}
 BCED & : \text{Rect}(BC, PQ), \\
 BMLD & : \text{Rect}(BM, PQ), \\
 MNKL & : \text{Rect}(MN, PQ), \\
 NCEK & : \text{Rect}(NC, PQ).
 \end{aligned}$$

The opposite-side congruences of the parallelograms are what transport the fixed side PQ through the two cuts. The conclusion still contains the same exact scissors equality between the concrete quadrilateral terms.

At this stage the geometry corresponding to Euclid’s wording is visible, but the theorem still uses one particular drawn rectangle and its particular cut points. The final step removes this representational dependence.

7 Representative Independence

7.1 Why representatives matter formally

A phrase such as

$$\text{Rect}(BC, PQ)$$

does not identify four vertices. Lean, however, must manipulate an actual term such as

```
hilbertParallelogramTerm Geo U0 U1 U2 U3
```

with named vertices. The final theorem therefore accepts arbitrary representatives for the whole rectangle and for each of the three part rectangles.

The concrete dissection first proves

$$BCED \simeq_{\text{sc}} BMLD + MNKL + NCEK.$$

Then four calls to

```
rectangle_contained_by_unique
```

transport the whole and the three pieces to arbitrary representatives U , V , W , and Z . Compatibility of `HilbertScissorsEq` with formal addition transports the nested sum. Transitivity completes the chain

$$\begin{array}{c} \text{arbitrary whole representative} \\ \Downarrow \\ \text{concrete whole rectangle} \\ \Downarrow \\ \text{concrete three-part dissection} \\ \Downarrow \\ \text{arbitrary representatives of the three parts.} \end{array}$$

This is the formal content of representative independence. It is not merely a convenience for Lean. It explains why the Euclidean expression “the rectangle contained by two segments” can function as a stable geometric magnitude without first quotienting segments into numerical lengths or assigning numerical areas to rectangles.

8 Commutativity by Rectangle Rotation

The second algebra-like feature is symmetry in the two containing segments. For a numerical product one would write $ab = ba$. The synthetic proof does not appeal to a commutative multiplication law. Instead it rotates the presentation of a rectangle.

If $ABCD$ is contained by PQ, RS , then the cyclically reoriented rectangle $BCDA$ has adjacent sides BC and CD . The first is already congruent to RS ; the second is congruent to AB , hence to PQ . Thus the same geometric rectangle, with its vertices read from the next corner, is contained by RS, PQ .

The exact theorem available in the rectangle layer is

```
rectangle_contained_by_swap
```

It returns both the swapped `IsRectangleContainedBy` fact and a `HilbertScissorsEq` identifying the two quadrilateral presentations. The current rectangle API also packages the representative-independent consequence as

```
rectangle_contained_by_comm
```

so the commutativity needed by later Book II propositions is a theorem of rectangle geometry rather than an assumed law of segment multiplication.

Figure 2 shows the geometric mechanism. The operation is a cyclic change of the distinguished corner of the same rectangle; this exchanges the two adjacent side roles.

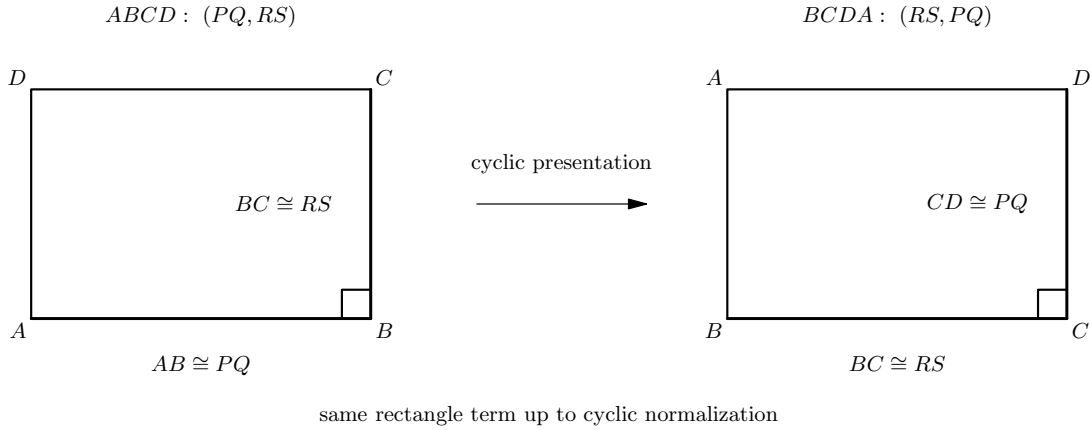


Figure 2: The geometric source of commutativity. On the left, $ABCD$ represents the rectangle contained by PQ, RS . Reading the same rectangle from the next vertex gives $BCDA$: now the first adjacent side represents RS and the second represents PQ . Exact scissors invariance under this cyclic presentation change yields the synthetic counterpart of exchanging the two factors.

This lemma is already operational in the next proposition. II.2 naturally writes rectangles in the order “whole side, part”, whereas the II.1 cut lemma produces them in the order “part, fixed side”. The implementation of II.2 uses the swap theorem to pass between these two presentations before applying the binary form of II.1.

9 Proof Architecture of the Reconstruction

The Lean proof is best read as five mathematical stages.

9.1 Step 1: exact binary dissection

```
rectangle_split_scissorsEq  
rectangle_split_left  
rectangle_split_right  
rectangle_split
```

The inherited I.47 split theorem supplies the exact scissors decomposition. The two local rectangle helpers prove that each piece remains a rectangle by propagating right angles through the relevant parallelograms and along the same rays determined by betweenness.

9.2 Step 2: iterate the split

```
rectangle_split_three
```

The right-hand remainder of the first split is itself a rectangle, so the same binary theorem can be applied again. Addition and transitivity of exact scissors equality compose the two decompositions.

9.3 Step 3: attach the “contained by” data

```
rectangle_split_three_contained_by
```

Opposite-side congruence transports the fixed segment PQ through all three pieces. Each concrete piece is thereby promoted from `IsRectangle` to `IsRectangleContainedBy`.

9.4 Step 4: remove the choice of representatives

```
rectangle_contained_by_unique  
euclid_proposition_2_1_three
```

The uniqueness theorem replaces the concrete whole and each concrete part by an arbitrary rectangle contained by the same pair of segments. This is the semantic step that makes the final statement faithful to Euclid’s definite phrase “the rectangle contained by.”

9.5 Step 5: expose the symmetric side order

```
rectangle_contained_by_swap  
rectangle_contained_by_comm
```

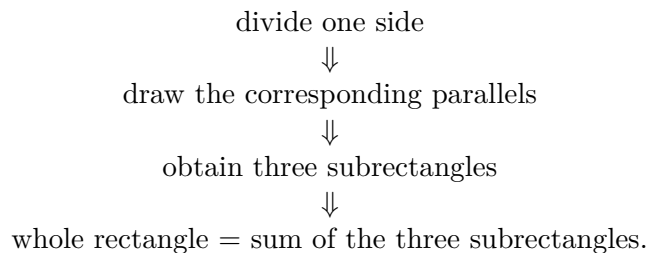
Cyclic rotation of the rectangle exchanges the two containing segments. This is not needed to establish the displayed three-part split itself, but it is an important part of the Book II rectangle interface and is immediately useful in II.2 and later propositions.

10 Classical Proof and Formal Proof

10.1 Euclid's proof pattern

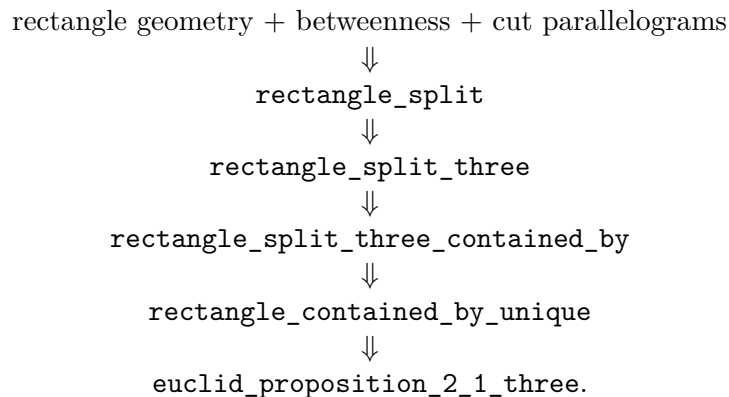
In the classical three-part configuration, take one undivided segment A and a segment BC divided at D and E . Erect a perpendicular at one endpoint of BC , lay off on it a segment equal to A , complete the outer rectangle, and through the division points draw parallels to the perpendicular side. The outer rectangle is visibly partitioned into three rectangles. Each smaller rectangle is contained by A and one of the three consecutive parts of BC . Common Notion 2 then combines the parts into the whole equality.

The classical DAG is therefore extremely small:



10.2 The formal DAG

The current reconstruction expands two things that the classical diagram treats as immediate: exact scissors bookkeeping and independence of the chosen rectangle representatives.



The proof is therefore not a mechanical transcription of Euclid. It preserves the geometric core, but makes the semantic and representational layers explicit.

11 Geometry, Content, and Representation

11.1 Geometry

The geometric data are substantial even though the proposition is visually simple. The current theorem receives the outer rectangle, strict betweenness relations on the divided side and its opposite, two diagonal-intersection witnesses, and four parallelogram facts for the two successive cuts. Right angles of the two resulting rectangles are not read from the picture; they are proved by right-angle transport and parallelogram geometry.

No coordinates are introduced. The cut points are controlled by **Between**, the side lengths by segment congruence, and the rectangular structure by parallels and synthetic right angles.

11.2 Content

Content enters in its strongest local form. The core statements are exact **HilbertScissorsEq** decompositions. The three-part theorem is composed from two exact binary splits by addition and transitivity. Representative transport is also exact scissors equality.

The more general equicomplementability relation is part of the surrounding Book I content theory, but II.1 does not need to weaken its conclusion to that level.

11.3 Representation

Representation is central to the formal meaning of the proposition. A rectangle contained by PQ, RS is not one canonical four-tuple of vertices. The predicate **IsRectangleContainedBy** describes a representative; **rectangle_contained_by_unique** proves that different representatives have the same exact scissors content.

There is a second representation issue inside the split itself. The inherited scissors theorem and the Book II statement use different cyclic orders for some parallelogram terms. The proof normalizes these by

```
parallelogram_term_rotateOne
```

before composing the scissors equalities. These cyclic changes are bookkeeping of a geometric representation, not new area facts.

12 Hidden Configuration Data

The classical drawing makes several facts appear automatic. Lean records them explicitly:

- $B - M - C$ for the first cut and $M - N - C$ for the second;
- $D - L - E$ and $L - K - E$ on the opposite side;
- $D - X - C$ together with $M - X - L$ for the first diagonal/cut intersection;
- $L - Y - C$ together with $N - Y - K$ for the second diagonal/cut intersection;
- explicit parallelogram witnesses for the left and right pieces of each split;
- congruence $CE \cong PQ$, which identifies the fixed side used in the “contained by” descriptions;
- four arbitrary rectangle representatives in the final theorem.

The points X and Y do not appear in Euclid’s statement and need not be emphasized in an elementary picture of II.1. They are nevertheless real data of the current exact-dissection API, inherited from the triangulated proof of the rectangle split.

13 Axiomatic Status

The public II.1 theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so II.1 is Euclidean in the current project API. Group IV is available through this typeclass and is consumed transitively by the parallelogram and parallel infrastructure on which the rectangle layer and inherited split theorem depend. The proposition itself does not introduce a new parallel axiom.

No continuity typeclass, Archimedean axiom, numerical area function, or segment field is introduced in `Proposition2_1.lean`. No proposition-local `axiom` or `sorry` occurs in the audited source. The modular builds for `HilbertRectangle` and `Proposition2_1` are recorded as clean at the present project checkpoint.

The content status is also worth separating from the later converse arguments of Book I. II.1 proceeds from geometry to an exact dissection. It does not infer metric geometry from an equal-content hypothesis, so it does not need a proper-part or De Zolt-type strictness argument.

14 Relationship with Book I and the Following Propositions

The exact split theorem reuses the scissors infrastructure established during I.47. In particular, the underlying theorem historically named

```
i47_square_split
```

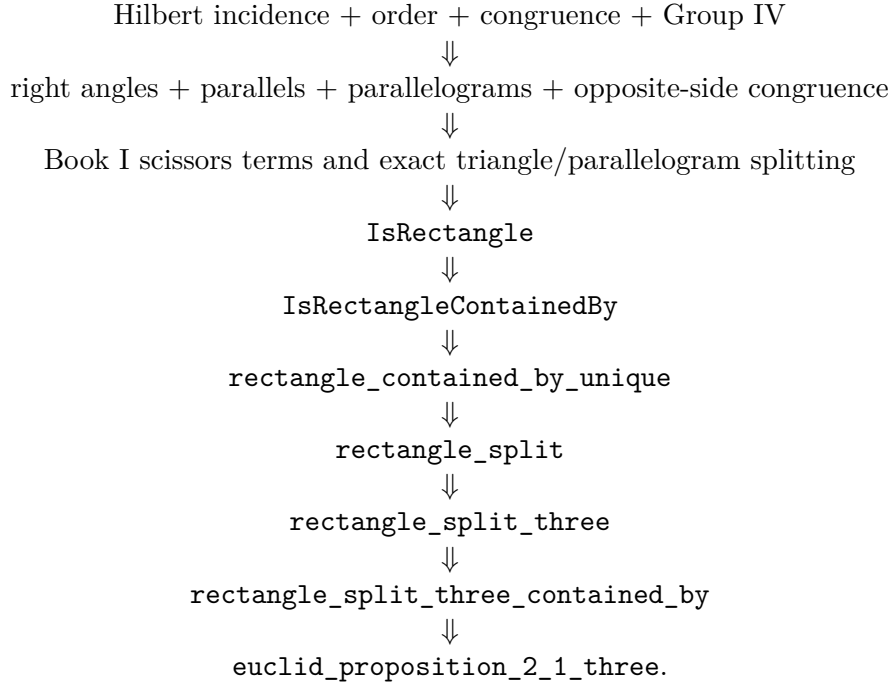
actually requires only parallelogram structure, and II.1 repackages it as a general rectangle split.

The rectangle layer also uses Book I results about parallelograms, opposite sides, congruent right angles, SAS/SSS transport, and the scissors representation of parallelograms. Thus Book II does not start a new foundational geometry; it starts a new way of organizing geometric magnitudes on top of the completed Book I infrastructure.

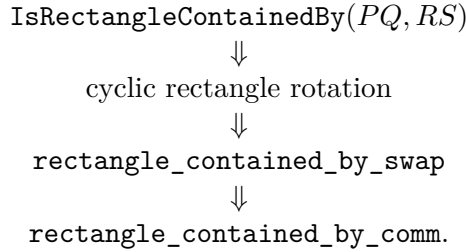
II.2 immediately imports II.1. Its formal proof illustrates why the symmetric rectangle API matters: Euclid's order of the two containing sides is opposite to the order produced by the binary cut theorem, so the representatives are swapped by rectangle rotation before II.1 is applied. II.3 continues the same rectangle calculus. In this sense II.1 is not only the first theorem of Book II but the API-level distributive kernel for the propositions that follow.

15 Dependency Summary

The actual formal dependency chain is



The symmetric side-order branch is



New geometric axiom in II.1:	no
New proposition-local axiom:	no
New rectangle abstraction:	yes, in <code>HilbertRectangle</code>
New reusable split helper:	yes
New representative-independence theorem:	yes
Exact scissors dissection used:	yes
General equicomplementability required by II.1:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Proposition-local <code>sorry/admit</code> :	no
Modular build:	clean

16 Conclusion

Proposition II.1 is the first clean instance in Book II of a product-like law being realized entirely by geometry. A segment is divided; a rectangle is cut along corresponding parallels; exact scissors equality identifies the whole with the sum of the pieces. The rectangle layer then removes the accidental choice of a concrete drawing, so Euclid’s phrase “the rectangle contained by” becomes a representative-independent synthetic magnitude.

The formalization also isolates the two structural laws that make the language look algebraic. Additivity comes from actual dissection. Commutativity comes from cyclic rotation of the rectangle. A square is the special case in which the two containing segments agree. None of these steps requires a product of segment lengths or a numerical area map.

Thus the “geometric algebra” of II.1 is best understood not as hidden modern symbolism but as a calculus of geometric magnitudes whose algebraic laws emerge from constructions, congruence, and finite decomposition.

The representative-independent three-part theorem at the end of the current formalization is:

```
theorem euclid_proposition_2_1_three
  [HilbertIncidence Geo]
  [HilbertEuclideanPlane Geo]

  -- Concrete dissection diagram.
  (B C D E L M K N X Y P Q : Geo.Point)
  (hRect : IsRectangle Geo B C E D)
  (hCE_PQ : Geo.Congruent C E P Q)

  -- First cut.
  (hBMC : Geo.Between B M C)
  (hDLE : Geo.Between D L E)
  (hDXC : Geo.Between D X C)
  (hMXL : Geo.Between M X L)
  (hLeftPar :
    IsParallelogram Geo L D B M)
  (hRightPar :
    IsParallelogram Geo C E L M)

  -- Second cut.
  (hMNC : Geo.Between M N C)
  (hLKE : Geo.Between L K E)
  (hLYC : Geo.Between L Y C)
  (hNYK : Geo.Between N Y K)
  (hMiddlePar :
    IsParallelogram Geo K L M N)
  (hFinalPar :
    IsParallelogram Geo C E K N)

  -- Arbitrary representatives of the four rectangles.
  (U0 U1 U2 U3 : Geo.Point)
  (V0 V1 V2 V3 : Geo.Point)
  (W0 W1 W2 W3 : Geo.Point)
  (Z0 Z1 Z2 Z3 : Geo.Point)

  (hWhole :
    IsRectangleContainedBy Geo
      U0 U1 U2 U3 B C P Q)
```

```

(hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B M P Q)

(hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 M N P Q)

(hPart3 :
  IsRectangleContainedBy Geo
    Z0 Z1 Z2 Z3 N C P Q) :
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo U0 U1 U2 U3)
  (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
    (hilbertParallelogramTerm Geo W0 W1 W2 W3 +
      hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)) := by

```